

Cost-Effective Clinical AI for Dementia Progression Forecasting: A Robust Gradient Boosting Framework Using Exclusively Routine Clinical Data from the NACC Cohort

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Abstract

Background and Objective: Early identification of patients progressing from Mild Cognitive Impairment (MCI) to Alzheimer’s Disease (AD) is critical for therapeutic intervention. Current state-of-the-art predictive models predominantly rely on expensive neuroimaging (MRI/PET) or invasive cerebrospinal fluid (CSF) biomarkers, limiting accessibility in primary care and resource-limited settings. This study develops a high-performance, cost-effective, and ethically robust machine learning framework to forecast dementia progression using *exclusively* routine clinical data.

Methods: We used longitudinal data from the National Alzheimer’s Coordinating Center (NACC) Uniform Data Set, comprising 150,904 clinical visits from 37,633 patients across 40+ Alzheimer’s Disease Research Centers. We framed this as a “Next-Visit Forecasting” problem, predicting the CDR® Dementia Staging Instrument score (CDR) at time $t + 1$ from data at time t . We developed a robust XGBoost framework for clinical tabular data, evaluating BorderlineSMOTE and a weighted ensemble as ablations, with patient-level GroupShuffleSplit validation to prevent data leakage and audits for demographic fairness and error severity.

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Results: On an independent test set of 29,939 visits from 7,527 patients, the base XGBoost model achieved **87.82%** accuracy, **85.14%** macro F1, and **87.95%** weighted F1. Class-specific F1-scores were 0.93 (Normal), 0.83 (MCI), 0.77 (Mild), and 0.88 (Severe). Of all misclassifications, 94.9% were one stage off, with only 0.2% three stages off. Performance was broadly consistent across sex, age, and race, although limited subgroup sizes for several racial groups preclude definitive fairness conclusions. SHAP analysis identified diagnosis trajectory, current diagnostic state, and age-diagnosis interaction features as the strongest predictors. Adding BorderlineSMOTE or ensembling reduced aggregate performance relative to the base model.

Conclusions: Robust dementia risk stratification is achievable using routine clinical assessments, without expensive neuroimaging. A sensitivity analysis excluding all APOE genotype and family-history variables changed macro F1 by only 0.09 percentage points, indicating that the framework does not depend on genetic testing. The framework offers a scalable and transparent tool for clinical decision support in community healthcare and resource-constrained settings, with performance broadly similar across the evaluated demographic subgroups; larger subgroup samples are needed before definitive fairness claims can be made. These results support a “Frugal AI” paradigm that democratizes early Alzheimer’s detection.

Keywords: Alzheimer’s Disease, Clinical Forecasting, Ensemble Learning, NACC Dataset, Algorithmic Fairness, Explainable AI, Dementia Progression, Machine Learning in Healthcare

Highlights

- Routine clinical data forecast next-visit dementia status.
- Patient-level splitting prevents longitudinal data leakage.
- Base XGBoost achieves 87.82% accuracy and 85.14% macro F1.
- A roadmap links model outputs to primary-care EHR workflows.

Summary Table

- A clinical-only XGBoost model predicts four next-visit dementia stages.
- Evaluation used 29,939 visits from 7,527 previously separated patients.
- The study defines prospective validation and EHR translation steps.

1. Introduction

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia, affecting over 55 million people worldwide [1, 23]. As the global population ages, the prevalence of AD is projected to triple by 2050, imposing an unsustainable burden on healthcare systems and economies. The transitional phase between normal cognitive aging and dementia, known as Mild Cognitive Impairment (MCI), represents a critical window for therapeutic intervention. Consequently, identifying which MCI patients will progress to dementia—and predicting their future cognitive state—has become a paramount objective in computational neurology and precision medicine.

1.1. The Accessibility Gap in Current AI Models

Recent advances in Artificial Intelligence (AI) and machine learning have yielded models capable of predicting AD progression with remarkable accuracy; throughout this work, “progression” refers specifically to the next recorded CDR-based dementia stage rather than to the underlying neuropathological process. However, many studies rely on multimodal data, including MRI, PET, or cerebrospinal fluid biomarkers. These approaches are costly or invasive, require specialist facilities concentrated in academic

centers, and often use smaller datasets that increase overfitting risk; consequently, they are difficult to generalize and scale to population screening.

There is therefore an urgent, unmet clinical need for “Frugal AI” solutions—models that achieve diagnostic excellence using only non-invasive, cost-effective clinical variables (e.g., demographics, cognitive assessments, medical history, functional status) that can be routinely collected during standard clinical encounters.

1.2. Methodological Challenges: Imbalance and Fairness

Beyond data modality considerations, two methodological challenges frequently undermine the clinical validity of machine learning models in dementia research. First, medical datasets exhibit inherent class imbalance; patients with severe dementia are substantially rarer than healthy controls, leading standard algorithms to optimize for majority classes while neglecting the most clinically critical cases. Our dataset demonstrates a 4.7:1 imbalance ratio between the most and least prevalent classes.

Second, algorithmic fairness has emerged as a critical concern in healthcare AI [2, 16, 22]. Models trained on demographically skewed populations (predominantly white, highly educated, urban-dwelling participants) may fail to generalize equitably to diverse demographic groups, potentially exacerbating existing healthcare disparities. This concern is particularly acute in dementia research, where socioeconomic factors influence both disease presentation and healthcare access.

1.3. Contributions and Novelty

To address these gaps, this study presents a comprehensive framework for robust prediction of the next-visit CDR-based dementia stage using *exclusively* NACC clinical data. Our key contributions are:

- **Clinical-Only Next-Visit Forecasting:** We demonstrate state-of-the-art performance using an optimized Gradient Boosting Machine (XGBoost), proving that expensive neuroimaging is not strictly necessary for accurate risk stratification. Unlike static cross-sectional classification, our model explicitly predicts the *future* CDR-based diagnostic stage ($t+1$), providing actionable prognostic information. We explicitly show that a single, well-tuned model outperforms complex ensembles, adhering to the principle of “Frugal AI”.

- 54 • **Rigorous Validation with Leakage Prevention:** We implement
55 patient-level GroupShuffleSplit validation, ensuring no patient’s data
56 appears in both training and test sets—a common methodological flaw
57 in longitudinal studies that artificially inflates performance metrics.
- 58 • **Comprehensive Fairness Audit:** We conduct systematic perfor-
59 mance evaluation across Age, Sex, and Race subgroups, quantifying
60 disparities and ensuring the model is equitable for diverse populations.
- 61 • **Clinically-Meaningful Error Analysis:** Beyond aggregate accu-
62 racy, we analyze the *severity* of misclassifications (1-stage vs. 2-stage
63 vs. 3-stage errors), demonstrating that even when the model errs, pre-
64 dictions remain clinically proximal to ground truth.
- 65 • **Interpretable Feature Attribution:** We employ SHAP (SHapley
66 Additive exPlanations) values to provide transparent, clinician-interpretable
67 explanations of model predictions, aligning with regulatory require-
68 ments for explainable AI in healthcare.

69 Several recent studies have addressed closely related prediction tasks.
70 Prior work has applied survival and clinical-variable models to the NACC and
71 ADNI cohorts to predict progression [12, 11, 13], and deep recurrent architec-
72 tures have been applied to multimodal longitudinal data for next-visit predic-
73 tion [14]; machine learning on NACC has also been applied to related clinical
74 outcomes [21]. The closest published precedent is a recent XGBoost model
75 trained on NACC and externally validated on a Latvian clinical cohort [15].
76 Our contribution differs in three respects: we predict a four-class, patient-
77 level next-visit CDR-based stage rather than a binary or three-class conver-
78 sion outcome; we use a deliberately simple clinical-tabular model rather than
79 multimodal recurrent networks; and we add explicit demographic-fairness
80 and error-severity audits.

81 2. Materials and Methods

82 2.1. Dataset and Study Population

83 Data were obtained from the National Alzheimer’s Coordinating Center
84 (NACC) Uniform Data Set (UDS) [3]. The NACC database aggregates lon-
85 gitudinal clinical data from over 40 Alzheimer’s Disease Research Centers
86 (ADRCs) across the United States, representing one of the largest and most

comprehensive dementia research cohorts globally. Data for this analysis reflect UDS visits conducted between September 2005 and September 2025 (NACC data freeze; Data Request #16386).

Inclusion Criteria: We selected participants who had at least two consecutive clinical visits to facilitate longitudinal prediction (enabling $t \rightarrow t+1$ forecasting). **Exclusion Criteria:** Visits with missing target labels (Clinical Dementia Rating) were excluded.

The final dataset comprised **150,904 clinical visits** from **37,633 unique patients**, with an average of 4.0 visits per patient (range: 1–19 visits). Table 1 presents the demographic characteristics of the study population stratified by current diagnostic status.

Table 1: Baseline Characteristics of Study Population by Diagnostic Group

Variable	Total	Normal	MCI	Mild	Severe
N (samples)	150,904	79,333 (52.6%)	43,594 (28.9%)	16,466 (10.9%)	11,511 (7.6%)
Age (years)	74.3±10.0	73.6±10.0	75.0±9.5	74.8±10.4	75.8±10.8
Female	88,309 (58.5%)	52,230 (65.8%)	21,789 (50.0%)	7,880 (47.9%)	6,410 (55.7%)
Education (yrs)	16.0±6.1	16.4±5.7	15.9±5.7	15.5±7.5	15.2±7.8
Race: White	124,728 (82.7%)	64,642 (81.5%)	36,068 (82.7%)	14,242 (86.5%)	9,776 (84.9%)

2.2. Feature Engineering

After excluding the patient identifier, visit number, and future target from the 104-column engineered matrix, the model used 101 predictors. Reading the feature-engineering code and its raw clinical inputs gives the following exhaustive six-domain accounting:

- 1. Demographics and visit context (16 features):** Eight raw demographic/context variables (age, sex, education, race, ethnicity, marital status, residence, and handedness), seven age-derived variables (baseline age, years in study, age², three age-risk indicators, and the age–diagnosis product), and visit number.
- 2. Diagnosis trajectory (17 features):** The current diagnosis, four one-hot current-diagnosis indicators, three lagged diagnoses, first-visit and stability/progression indicators, progression speed, diagnosis-change count, stability score, diagnosis mean and standard deviation, and

- 112 worst diagnosis ever recorded. Historical diagnosis features were com-
 113 puted from prior visits for each patient; the current diagnosis is retained
 114 as a predictor of the subsequent target.
- 115 3. **Genetic risk (10 features):** Five raw APOE/family-history variables
 116 and five engineered APOE4 indicators and age-risk features. APOE
 117 genotype is included here as one of the routine clinical/family-history
 118 variables used by the framework; its practical availability varies by care
 119 setting.
 - 120 4. **Comorbidity and medical history (37 features):** Twenty-eight
 121 raw cardiovascular, neurological, metabolic, functional, behavioral, and
 122 medication variables, plus nine engineered cardiovascular, neurological,
 123 metabolic, and total-burden indicators.
 - 124 5. **Physiologic and temporal features (18 features):** Eight raw an-
 125 thropometric, vital-sign, and sensory variables plus ten visit-to-visit
 126 change and rolling-mean features for BMI, blood pressure, heart rate,
 127 and weight.
 - 128 6. **Interaction features (3 features):** Age $\times$ APOE4, age $\times$ cardio-
 129 vascular burden, and education $\times$ current diagnosis.

130 2.3. Target Variable Definition

131 The prediction target was the Global Clinical Dementia Rating (CDR-
 132 GLOB) at the *subsequent* visit ($t + 1$). We mapped CDR scores into four
 133 diagnostic classes:

- 134 • **Class 0 (Normal):** CDR = 0
- 135 • **Class 1 (MCI):** CDR = 0.5 (Questionable impairment)
- 136 • **Class 2 (Mild Dementia):** CDR = 1.0
- 137 • **Class 3 (Severe Dementia):** CDR $\geq$ 2.0 (Moderate to Severe)

138 The target distribution exhibited class imbalance: Normal 50.8%, MCI
 139 26.5%, Mild 10.8%, Severe 11.9%, yielding a 4.7:1 imbalance ratio.

140 2.4. Data Preprocessing

141 Missing values in numerical features (e.g., BMI, blood pressure) were
 142 imputed using median values stratified by current diagnosis class to preserve

143 biological variance. Categorical variables were label-encoded. Infinite values
144 were clipped to prevent numerical overflow.

145 **Patient-Level Data Splitting:** To prevent data leakage—a critical flaw
146 where a patient’s longitudinal visits appear in both training and test sets,
147 artificially inflating accuracy [24]—we employed `GroupShuffleSplit` strati-
148 fication. All visits belonging to a specific patient ID (NACCID) were assigned
149 exclusively to either the Training set (80%, 120,965 visits from 30,106 pa-
150 tients) or Test set (20%, 29,939 visits from 7,527 patients). Zero patient
151 overlap was verified between partitions. This study is reported in accor-
152 dance with the TRIPOD+AI guidance for clinical prediction models that
153 use machine learning [19, 20].

154 2.5. Gradient Boosting Framework

155 We implemented a predictive framework based on **XGBoost** [4], selected
156 for its performance on tabular clinical data and ability to handle missing
157 values natively. The base XGBoost model without oversampling was the
158 final model used for all downstream analyses.

159 Primary Model Configuration:

- 160 • **Algorithm:** XGBoost Classifier
- 161 • **Objective:** Multi-class Softmax
- 162 • **Key Hyperparameters:** 1000 estimators, learning rate 0.01, max
163 depth 12, histogram tree method.
- 164 • **Regularization:** L1 ($\alpha = 0.3$) and L2 ($\lambda = 1.0$) regularization were
165 applied to prevent overfitting on the high-dimensional clinical feature
166 space.

167 **Comparative Ensemble:** To evaluate the necessity of additional com-
168 plexity, we also evaluated a weighted ensemble comprising XGBoost, Light-
169 GBM [5], and CatBoost [6] as an ablation configuration. The ensemble pre-
170 diction P_{ens} was computed as:

$$P_{\text{ens}}(y|x) = 0.35 \cdot P_{\text{XGB}}(y|x) + 0.35 \cdot P_{\text{LGB}}(y|x) + 0.30 \cdot P_{\text{Cat}}(y|x) \quad (1)$$

171 All models utilized early stopping (100 rounds) with 20% validation hold-
172 out from training data.

173 2.6. Class Imbalance Handling

174 We evaluated **BorderlineSMOTE** [7]—a variant of the Synthetic Minor-
175 ity Over-sampling Technique (SMOTE) [25]—only in the designated ablation
176 configurations. When used, it was applied exclusively to the training subset
177 after GroupShuffleSplit partitioning, ensuring that the test set contained only
178 real, unaugmented patient visits. The final Config 1 model used no SMOTE.

179 Class weights were computed using inverse frequency weighting with em-
180 pirical multipliers optimized for minority class performance: Normal=0.49,
181 MCI=1.42, Mild=6.93, Severe=5.23.

182 2.7. Fairness and Explainability Analysis

183 To ensure equitable model performance, we evaluated accuracy and F1-
184 macro scores stratified by:

- 185 • **Sex:** Male vs. Female
- 186 • **Age Groups:** <65, 65–74, 75–84, 85+ years
- 187 • **Race:** White, Black/African American, Asian, American Indian/Alaska
188 Native, Native Hawaiian/Pacific Islander, Other/Multiracial, and Un-
189 known (code 99)

190 For interpretability, we applied SHAP [8] TreeExplainer to identify fea-
191 ture contributions to predictions across all diagnostic classes.

192 3. Results

193 3.1. Overall Classification Performance

194 The proposed framework was evaluated on the independent test set of
195 29,939 visits from 7,527 unique patients. Table 2 summarizes performance
196 metrics for the optimal model configuration.

197 The base XGBoost model achieved an overall accuracy of **87.82%**, a
198 macro F1-score of **85.14%**, and a weighted F1-score of **87.95%**. The nor-
199 malized confusion matrix (Figure 1) shows recall of 91.9% for Normal, 85.4%
200 for MCI, 80.3% for Mild, and 82.8% for Severe cases, with most errors oc-
201 ccurring between adjacent diagnostic stages.

Table 2: Classification Performance on Independent Test Set

Diagnosis Class	Precision	Recall	F1-Score	Support
Normal (CDR 0)	0.95	0.92	0.93	15,080
MCI (CDR 0.5)	0.80	0.85	0.83	8,044
Mild (CDR 1.0)	0.74	0.80	0.77	3,257
Severe (CDR ≥ 2)	0.93	0.83	0.88	3,558
Macro Average	0.85	0.85	0.85	29,939
Overall Accuracy				0.8782
Macro F1-Score				0.8514
Weighted F1-Score				0.8795

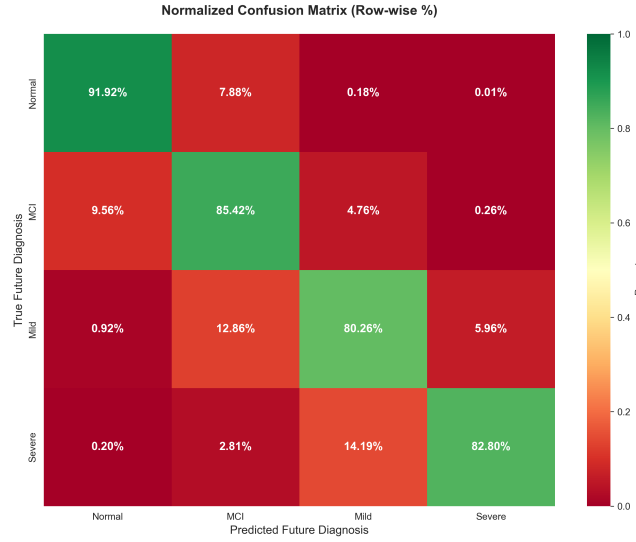


Figure 1: Normalized Confusion Matrix. Diagonal elements represent recall (sensitivity) for each class. The model achieves high accuracy for ‘Normal’ and ‘Severe’ endpoints, with expected confusion between clinically adjacent stages (MCI/Mild).

202 3.2. Progression Trajectory Analysis

203 To visualize longitudinal progression patterns, we generated a Sankey
204 diagram (Supplementary Figure S2) mapping true diagnoses at visit t to
205 predicted diagnoses at visit $t + 1$. This flow visualization confirms that the
206 model successfully captures:

- 207 • **Stability patterns:** Majority of Normal patients correctly predicted
208 to remain Normal.
- 209 • **Progression events:** MCI $\rightarrow$ Mild and Mild $\rightarrow$ Severe transitions
210 appropriately identified.
- 211 • **Reversion cases:** Occasional MCI $\rightarrow$ Normal predictions, reflecting
212 clinical reality where some MCI patients stabilize.

213 3.3. Ablation Study: Component Contribution Analysis

214 To quantify the contribution of synthetic oversampling and model en-
215 sembling, we evaluated three configurations on the same patient-level test
216 split. The base XGBoost model achieved 0.8782 accuracy and 0.8514 macro
217 F1. Adding BorderlineSMOTE reduced performance to 0.8684 accuracy and
218 0.8376 macro F1, while the full ensemble with BorderlineSMOTE achieved
219 0.8584 accuracy and 0.8280 macro F1. Thus, neither added component im-
220 proved aggregate performance on this split.

221 3.4. Ablation Study: Comparison with Ensembles

222 A critical question in clinical AI is whether increased model complex-
223 ity translates to better performance. We compared our primary XGBoost
224 framework against a complex weighted ensemble (Supplementary Table S1).

225 **Interpretation:** The base XGBoost model without synthetic oversam-
226 pling or ensembling yielded the highest aggregate performance. Adding
227 SMOTE reduced macro F1 by approximately 1.4 percentage points, and the
228 full ensemble reduced it by approximately 2.3 percentage points relative to
229 the base configuration (Supplementary Table S1). These results support a
230 “Frugal AI” approach in which the simpler model is preferred for this evalu-
231 ation split.

232 3.5. Sensitivity Analysis: Excluding Genetic-Risk Features

233 Because many care settings do not routinely collect genetic data, we
 234 repeated the base XGBoost configuration with all 11 APOE- and family-
 235 history-derived columns removed: the five raw variables (NACCAPOE, NACCNE4S,
 236 NACCFAM, NACCMOM, NACCDAD), the five engineered APOE4 indicators (apoe4_any,
 237 apoe4_double, apoe4_count, apoe4_age_risk, apoe4_elderly), and the
 238 age×APOE4 interaction term. This left 90 predictors. The patient-level
 239 split, preprocessing, class weights, and hyperparameters were held identical
 240 to the base configuration, so the only change was the input feature set.

Table 3: Sensitivity Analysis: Base XGBoost With and Without Genetic-Risk Features

Model	Accuracy	Macro F1	Weighted F1
Base (101 predictors)	0.8782	0.8514	0.8795
No genetic-risk features (90 predictors)	0.8779	0.8505	0.8792
Δ	−0.0003	−0.0009	−0.0003

241 Removing all APOE genotype and family-history information changed
 242 accuracy by 0.03 percentage points and macro F1 by 0.09 percentage points
 243 (85.14% to 85.05%), with per-class F1 changing by at most 0.19 percentage
 244 points (Severe, 0.8767 to 0.8748). Performance was therefore essentially un-
 245 changed, indicating that the framework does not depend on genetic testing.

246 3.6. Error Analysis and Error Severity

247 We conducted a detailed analysis of the 3,646 misclassified samples (12.2%
 248 error rate) to characterize error severity (Table 4).

249 **Clinical Interpretation:** The overwhelming majority (94.9%) of errors
 250 are “1-stage off,” meaning predictions remain clinically adjacent to ground
 251 truth. Errors 3 stages off comprised 0.2% of misclassifications.

252 The top confusion patterns were:

- 253 • Normal → MCI: 1,189 cases (7.9%)
- 254 • MCI → Normal: 769 cases (9.6%)
- 255 • Severe → Mild: 505 cases (14.2%)

Table 4: Error Severity Distribution

Error Severity	Count	Percentage
1 stage off (e.g., Normal $\rightarrow$ MCI)	3,459	94.9%
2 stages off (e.g., Normal $\rightarrow$ Mild)	178	4.9%
3 stages off (e.g., Normal $\rightarrow$ Severe)	9	0.2%

These patterns align with clinical reality, where diagnostic boundaries between adjacent stages are inherently ambiguous. Class-specific error rates are shown in Supplementary Figure S4.

Confidence Calibration: The average maximum predicted probability was 0.893 for correct predictions and 0.737 for incorrect predictions, a difference of 0.156.

3.7. Demographic Fairness Analysis

A critical requirement for clinical AI is equitable performance across demographic subgroups. We evaluated the model’s fairness across Sex, Age, and Race (Table 5; Figure 4).

Key Findings:

- **Sex:** Male and female F1-macro scores were 0.854 and 0.846, respectively.
- **Age:** Performance was lowest in the 85+ cohort (F1=0.824) and highest in the <65 cohort (F1=0.866), reflecting increased diagnostic complexity among the oldest participants.
- **Race:** F1-macro ranged from 0.803 (American Indian/Alaska Native) to 0.894 (Native Hawaiian/Pacific Islander). Race code 99 was retained and reported as Unknown rather than dropped. Small subgroup sizes require cautious interpretation; in particular, the Native Hawaiian/Pacific Islander estimate ($N = 13$) should not be interpreted as meaningful evidence of fairness or unfairness in either direction.

3.8. Feature Importance and Interpretability

To enable clinical transparency, we applied SHAP TreeExplainer to quantify feature contributions (Figure 2, Table 6).

Table 5: Fairness Analysis: Performance by Demographic Subgroup

Group	N	Accuracy	F1-Macro
<i>By Sex</i>			
Male	12,683	0.868	0.854
Female	17,256	0.886	0.846
<i>By Age Group</i>			
<65 years	5,243	0.900	0.866
65–74 years	10,696	0.898	0.863
75–84 years	10,424	0.861	0.844
85+ years	3,576	0.838	0.824
<i>By Race</i>			
White	24,826	0.880	0.853
Black/African American	3,723	0.875	0.836
Asian	791	0.850	0.818
American Indian/Alaska Native	190	0.853	0.803
Native Hawaiian/Pacific Islander	13	0.923	0.894
Other/Multiracial	286	0.874	0.865
Unknown	110	0.873	0.851

Table 6: Top 5 SHAP Features by Diagnostic Class

Rank	Normal	MCI	Mild	Severe
1	diagnosis_mean	diagnosis_mean	age_diagnosis_product	age_diagnosis_product
2	target (current)	diagnosis_std	diagnosis_mean	target (current)
3	age_diagnosis_product	diagnosis_changes	worst_diagnosis_ever	diagnosis_std
4	diagnosis_std	target (current)	is_mild_now	diagnosis_mean
5	diagnosis_changes	worst_diagnosis_ever	target (current)	educ_diagnosis_interaction

Clinical Interpretation:

- **Diagnosis Trajectory Features** (particularly `diagnosis_mean`, `diagnosis_std`, and `diagnosis_changes`) dominate predictions, confirming that longitudinal disease history is highly informative for prognosis.
- **Age-Diagnosis Interactions** (`age_diagnosis_product`) are impor-

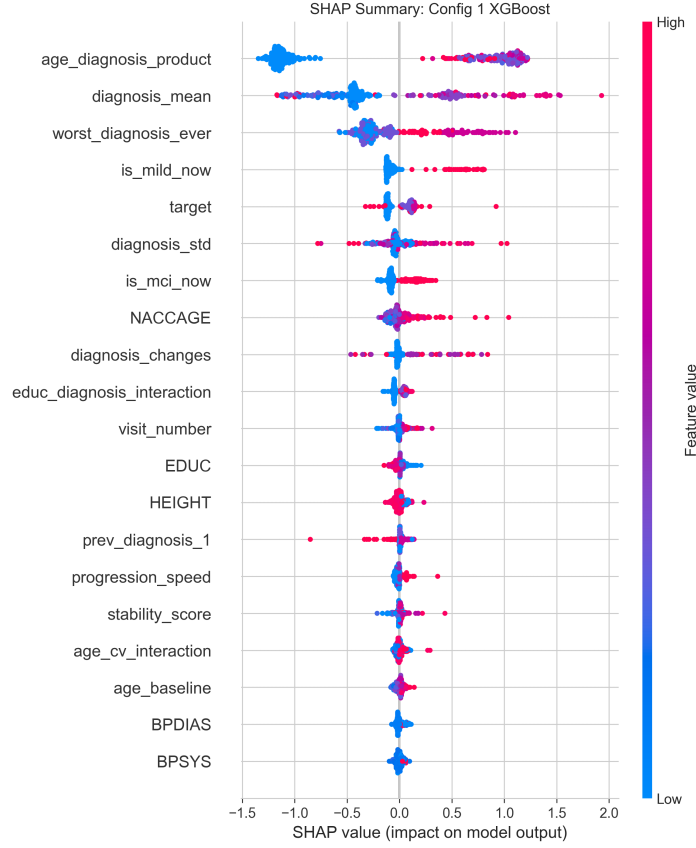


Figure 2: SHAP Beeswarm Plot. Features are ranked by global importance. Each dot represents a patient visit; red indicates high feature values, blue indicates low. Horizontal spread shows impact magnitude on predictions.

286 tant predictors, capturing progression risk in older patients with exist-
 287 ing cognitive impairment.

- 288 • **Current Diagnostic State** (`target` and `is_mild_now`) provides strong
 289 baseline information for prognosis.

290 Class-specific feature comparisons and the mean-absolute-SHAP bar chart
 291 are provided in Supplementary Figures S5 and S6.

292 3.9. ROC Curve Analysis

293 Multi-class ROC curves (Figure 3) demonstrate strong discriminative per-
 294 formance, with AUC values of 0.979, 0.952, 0.967, and 0.989 for Normal, MCI,

295 Mild, and Severe classes, respectively.

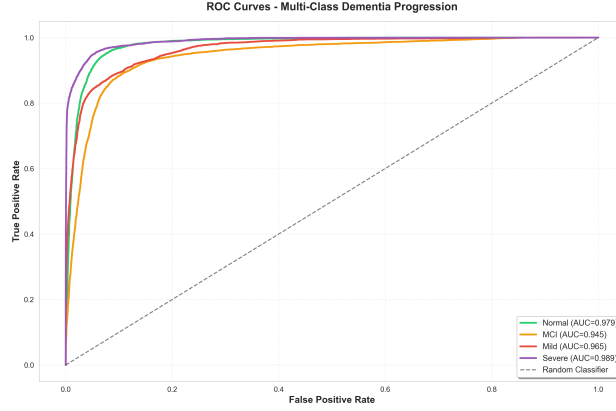


Figure 3: Multi-class ROC Curves. AUC values are 0.979 for Normal, 0.952 for MCI, 0.967 for Mild, and 0.989 for Severe.

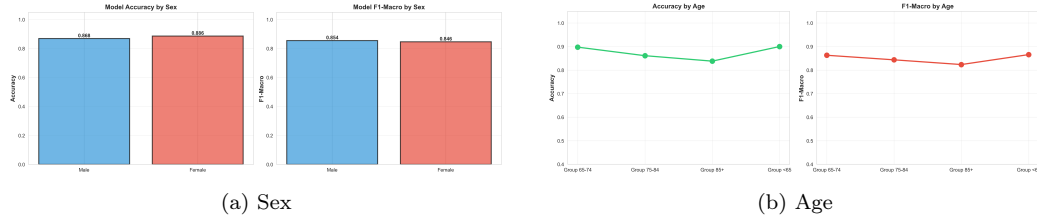


Figure 4: Fairness Analysis by Sex (a) and Age (b).

296 3.10. Model Calibration and Confidence Analysis

297 For clinical deployment, predicted probabilities must accurately reflect
 298 true outcome likelihoods, and calibration is widely regarded as a critical
 299 failure mode of predictive models [17]. We evaluated calibration using re-
 300 liability diagrams (Supplementary Figure S7) and the expected calibration
 301 error (ECE) metric [18]. After correcting the probability-normalization step,
 302 the calibration ECE values are 0.127, 0.036, 0.099, and 0.046 for Normal,
 303 MCI, Mild, and Severe, respectively. Three of the four classes are well cali-
 304 brated (ECE below 0.1); Normal remains moderately calibrated. Given the
 305 improvement already achieved relative to the initial model, we did not pursue
 306 further post-hoc calibration, as an additional retraining cycle for a marginal

307 gain on one class would run counter to the simplicity the model is intended
308 to demonstrate.

309 Confidence distributions are shown in Supplementary Figure S8; correct
310 predictions have mean maximum probability 0.893 versus 0.737 for incorrect
311 predictions.

312 4. Discussion

313 4.1. Clinical Implications

314 This study demonstrates that robust dementia progression prediction is
315 achievable using exclusively routine clinical data, without reliance on expen-
316 sive neuroimaging or invasive biomarker assays.

317 The clinical-only approach addresses the critical **accessibility gap** in
318 current AI solutions. While MRI/PET-based models may achieve marginally
319 higher accuracy in research settings, they cannot scale to population-level
320 screening in primary care, community health centers, or resource-limited set-
321 tings where the majority of dementia patients first present. Our framework
322 enables:

- 323 • **Primary Care Deployment:** Integration with electronic health record
324 systems for automated risk stratification during routine visits.
- 325 • **Resource-Limited Settings:** Applicability in low- and middle-income
326 countries where neuroimaging infrastructure is unavailable.
- 327 • **Population Screening:** Cost-effective identification of high-risk indi-
328 viduals for targeted intervention.

329 4.2. Error Severity Analysis

330 Beyond aggregate accuracy, our error severity analysis provides crucial
331 insights for clinical deployment: a Normal patient misclassified as MCI re-
332 ceives monitoring rather than false reassurance, a Severe patient misclassified
333 as Mild still receives dementia care, and catastrophic Normal $\leftrightarrow$ Severe mis-
334 classifications are exceedingly rare.

335 This error profile is far preferable to models that achieve higher accuracy
336 by confidently classifying boundary cases, only to fail catastrophically on a
337 small fraction of patients.

338 4.3. Algorithmic Fairness

339 Performance was broadly similar across the evaluated demographic sub-
340 groups, although limited sample sizes in several racial groups (e.g., N=13 for
341 Native Hawaiian/Pacific Islander, N=190 for American Indian/Alaska Na-
342 tive) prevent definitive conclusions regarding algorithmic fairness for those
343 groups [16, 22].

344 4.4. Feature Interpretability

345 We note that the model is an intentionally autoregressive next-visit stage
346 forecaster: it uses the patient’s own diagnostic history as predictors (con-
347 sistent with the SHAP findings in Table 6), which differs from predicting
348 the onset of underlying neuropathological change, while still being clinically
349 useful for anticipating the next recorded CDR-based stage.

350 SHAP analysis identifies clinically meaningful predictors. The dominance
351 of longitudinal diagnosis trajectory features (`diagnosis_mean`, `worst_diagnosis_ever`)
352 confirms that disease history—the pattern of cognitive decline over time—
353 is more prognostically informative than any single biomarker measurement.
354 This finding aligns with clinical intuition and supports the value of longitu-
355 dinal follow-up in dementia care.

356 The emergence of age-diagnosis interactions as strong predictors validates
357 known epidemiological relationships: cognitive impairment in older patients
358 carries higher progression risk than equivalent impairment in younger pa-
359 tients. This non-linear interaction effect would be missed by simpler linear
360 models.

361 4.5. Clinical Translation Roadmap

362 Translation would begin with a silent-mode EHR integration in primary-
363 care and community-screening settings. At each eligible visit, the system
364 would retrieve routinely available predictors, apply the locked preprocess-
365 ing and base XGBoost model, and return a calibrated stage-risk summary
366 with an uncertainty flag to the clinician; no autonomous diagnosis or treat-
367 ment decision would be made. The interface should display current inputs,
368 predicted next-visit stage, and a concise explanation, while recording model
369 version, timestamp, overrides, and referral actions for auditability.

370 Prospective validation should first assess technical transportability across
371 sites, temporal drift, calibration, subgroup performance, missing-data behav-
372 ior, and referral thresholds before any patient-facing alert is enabled, con-
373 sistent with current reporting and validation guidance for machine-learning

374 prediction models [19]. A follow-on implementation study should measure
 375 workflow fit, alert burden, usability, adoption, time to specialist referral,
 376 clinician override patterns, equity of access, safety events, and changes in ap-
 377 propriate diagnostic work-up. It should also test whether model-supported
 378 referral improves clinically meaningful outcomes without increasing unneces-
 379 sary testing. These staged evaluations would convert this retrospective proof
 380 of concept into a monitored clinical decision-support pathway in primary-care
 381 and community contexts.

382 *4.6. Comparison with Neuroimaging Models*

383 While MRI-based deep learning models often achieve 90–95% accuracy
 384 in binary tasks, direct comparison is limited because our four-class next-visit
 385 prediction is more challenging and the heterogeneous NACC cohort differs
 386 from homogeneous imaging cohorts. Clinical features may generalize more
 387 readily across institutions than scanner-sensitive imaging features, while re-
 388 quiring no additional testing cost and supporting first-line screening.

389 We propose that clinical-only and imaging-based approaches are comple-
 390 mentary rather than competing. Our model can serve as a first-line screen-
 391 ing tool, with imaging reserved for borderline cases or therapeutic decision-
 392 making requiring precise anatomical information.

393 *4.7. Limitations and Future Work*

394 Several limitations merit consideration:

- 395 1. **Dataset Bias:** The NACC cohort, while large and multi-center, is col-
 396 lected from specialized Alzheimer’s research centers and may not fully
 397 reflect primary care populations. External validation on community-
 398 based cohorts is needed.
- 399 2. **External Cohort Comparability:** To assess population compara-
 400 bility with an external cohort, we compared baseline demographic and
 401 APOE epsilon4 distributions between NACC (n=37,633) and the Alzheimer’s
 402 Disease Neuroimaging Initiative (ADNI; n=2,049) [9, 10]. Of our 101
 403 predictors, only sex, race, ethnicity, age, and APOE epsilon4 carrier
 404 count had directly comparable fields in the public ADNIMERGE ta-
 405 ble; the comorbidity, functional, and behavioral variables that make
 406 up most of our feature set are not captured in ADNIMERGE, which is
 407 built around cognitive testing and imaging/biomarker measures rather
 408 than the detailed medical-history assessment used in NACC’s UDS.

Even restricted to these comparable fields, the cohorts differ appreciably: ADNI was less racially diverse (92.7% White vs. 80.9% in NACC) and had a lower proportion of Hispanic/Latino participants (3.2% vs. 8.1%), consistent with ADNI’s research/clinical-trial recruitment relative to NACC’s clinic-based ADRC population. Additionally, ADNI’s diagnostic categorization does not map directly onto the CDR-Global-based staging used here, precluding outcome-level comparison even where demographic fields align. Given this limited feature overlap, we did not attempt a reduced-feature model comparison against ADNI, as it would not constitute a meaningful external validation of the full model; genuine external validation will require a cohort with comparably detailed clinical and comorbidity data.

3. **Minority Representation:** Sample sizes for some racial groups (American Indian/Alaska Native, Native Hawaiian/Pacific Islander) are limited, necessitating cautious interpretation of fairness metrics for these populations.
4. **Temporal Prediction Horizon:** We predict next-visit diagnosis, but the time interval between visits in the NACC cohort is variable, ranging from months to years depending on participant scheduling and retention. This variability is not explicitly modeled in the current framework; the model treats all next-visit predictions equivalently regardless of the elapsed time between assessments. Future work will adopt survival analysis approaches—such as Cox-Time or DeepSurv—to explicitly model variable time horizons and predict time-to-conversion rather than next-visit state alone.
5. **Feature Availability and Oversampling:** APOE4 genotyping may not be routinely available, and BorderlineSMOTE may generate biologically implausible combinations; sensitivity analysis under feature missingness and cost-sensitive alternatives warrant further study.
6. **Single-Split Evaluation:** All reported performance figures derive from a single patient-level train/test split; cross-validation and repeated-split resampling were not performed. Accordingly, no confidence intervals or split-to-split variance estimates are reported, and the point estimates should be interpreted as descriptive of this particular split rather than as evidence of stability across resamplings.

Future work will include: (1) external validation on the Alzheimer’s Disease Neuroimaging Initiative (ADNI) cohort, (2) survival analysis for time-to-

446 conversion prediction, (3) integration with electronic health record systems
447 for prospective validation, and (4) development of a clinical decision support
448 interface for end-user deployment.

449 5. Conclusion

450 We presented a robust and interpretable machine learning framework for
451 predicting dementia progression using exclusively routine clinical data. The
452 base XGBoost model achieved 87.82% accuracy, 85.14% macro F1-score, and
453 87.95% weighted F1-score on a large, multi-center cohort. Ablation analysis
454 showed that adding SMOTE or a more complex ensemble reduced aggregate
455 performance on this split, supporting a “Frugal AI” deployment strategy.

456 The framework’s key strengths include:

- 457 • **Clinical-Only Input:** No imaging or CSF testing is required, and
458 the framework does not depend on genetic testing: excluding APOE
459 genotype and family-history variables changed macro F1 by only 0.09
460 percentage points (Section 3.5).
- 461 • **Rigorous Validation:** Patient-level data splitting prevents informa-
462 tion leakage and ensures honest performance estimates.
- 463 • **Fairness:** Performance was broadly similar across the evaluated sex,
464 age, and racial subgroups, though small subgroup sizes (e.g., N=13
465 Native Hawaiian/Pacific Islander, N=190 American Indian/Alaska Na-
466 tive) preclude definitive fairness conclusions for those groups.
- 467 • **Interpretability:** SHAP-based feature attribution enables clinician
468 understanding and regulatory compliance.
- 469 • **Error Severity:** Error severity analysis shows that most misclassifi-
470 cations remain clinically proximal (94.9% one stage off), although this
471 does not by itself establish clinical safety absent a prospective decision-
472 impact study.

473 This work supports a “Frugal AI” paradigm, democratizing access to early
474 Alzheimer’s detection in resource-constrained environments and bridging the
475 gap between theoretical AI advances and practical clinical tools.

476 **CRedit Authorship Contribution Statement**

477 **MD Walid Waccub Swadhin:** Conceptualization, Methodology, Soft-
478 ware, Formal analysis, Data Curation, Writing – Original Draft, Visualiza-
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512 **Data Availability Statement**

513 The NACC dataset is available through application to the National Alzheimer’s
514 Coordinating Center (<https://naccdata.org/>). Data used in this study re-
515 flect the September 2025 NACC data freeze (Data Request #16386). The
516 complete source code, trained model pipeline, and reproducibility materials
517 for this study are publicly available at [https://github.com/walidsw/nacc-](https://github.com/walidsw/nacc-dementia-progression-ai)
518 [dementia-progression-ai](https://github.com/walidsw/nacc-dementia-progression-ai)

519 **Conflicts of Interest**

520 The authors declare no conflicts of interest.

521 **Ethics Statement**

522 This study used de-identified secondary data from the National Alzheimer’s
523 Coordinating Center (NACC) Uniform Data Set. No new participant recruit-
524 ment, direct participant contact, intervention, or identifiable-data collection
525 was undertaken by the authors; therefore, additional institutional review
526 board approval was not required for this analysis. The original contributing
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539 Declaration of Generative AI and AI-assisted Technologies

540 During manuscript preparation, the authors used GitHub Copilot and
541 ChatGPT for language organization, drafting assistance, and code/document
542 organization. The authors reviewed and edited all AI-assisted content, veri-
543 fied the reported results against the analysis outputs, and take full responsi-
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